

Incorporating Health Cobenefits in Decision-Making for the Decommissioning of Coal-Fired Power Plants in China

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Cite This: <https://dx.doi.org/10.1021/acs.est.0c03310>



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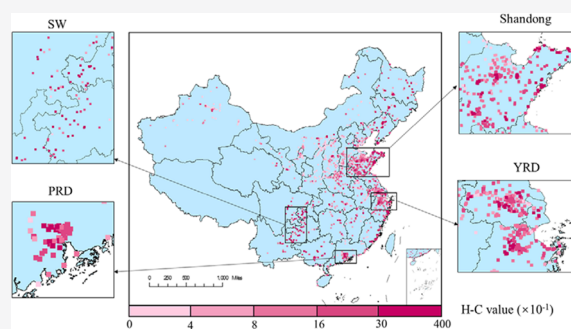


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Supporting Information

ABSTRACT: China's coal-fired power industry urgently needs deep decarbonization to meet the challenge of climate change. Regional air quality improvement and the health benefits can motivate efforts to achieve low-carbon goals. However, the health cobenefit per amount of carbon reduction may vary drastically across power plant units. The strategy of targeting more health cobenefits has been considered in designing an efficient carbon mitigation pathway, whereas this issue has not been analyzed at the unit level. In this study, an indicator called health benefit by carbon reduction (H/C) was constructed for each power unit to assess the relative potential of obtaining health cobenefits. The results reveal that the distribution of H/C values among units is extremely uneven: the first 1, 5, and 20% of the total carbon emission contributed to nearly 20, 40, and 70%, respectively, of the total health effects. The additional health benefits from H/C optimization were evaluated, and the decommissioning pathway of China's coal-fired power industry for achieving more health benefits was explored.



The H-C indicator: the health effects per amount of carbon emission, or the potential health benefit per amount of carbon reduction.

INTRODUCTION

Attainment of the 1.5 °C target proposed in the Paris Agreement requires near-zero or even negative carbon emissions in 2050.^{1–3} China's power generation industry, especially coal-fired power plants, contributed to a large amount of carbon emissions and urgently needs deep decarbonization. As energy infrastructures, the committed carbon emission of power plants will also hinder the realization of climate goals.⁴ Previous studies disclosed that climate goals require that the existing China's coal-fired power plants should be phased out before their life expectancy.⁵

Climate action is often hampered by the lagging effect of its direct benefits. To effectively boost climate actions, the benefit of air pollutant coabatement has been incorporated into the cobenefit assessments of climate policies in recent studies.^{6–16} Previous studies focused on China's power industry and declared that the air quality improvement of carbon measures, especially the regional health cobenefit, may offset or even exceed the mitigation costs.¹⁷ Among all air pollutants, ambient PM_{2.5} has the greatest effect on human health.

Furthermore, some more recent studies disclosed that health cobenefits resulting from the same amount of carbon reduction may vary drastically by region or the characteristics of power plant units.^{18–21} Goodkind et al.²² claimed that 10% of total air pollution emissions concentrated in or near the densely populated areas could account for 40% of the total damage. Tong et al.²³ estimated that closing some specific coal-fired

power plants, which account for 0.8% of total capacity, could reduce 16% of the total primary PM_{2.5} emissions in China. Therefore, how to promote the attainment of health cobenefits through climate action needs to be valued. This issue was explored at the grid level in our previous study,¹⁵ where we found that the health cobenefits per carbon reduction among grids may vary as much as fivefold. Nevertheless, analyses at the grid level could only provide limited differentiated information and regional strategies. With the continuous development of China's unit-level coal-fired power data set,^{23–26} this issue could be investigated thoroughly at the unit level to provide more specific suggestions.

Recently, substantial air pollution emission reductions have occurred among China's power plants due to the implementation of ultra-low emission renovations (ULERs),^{27–30} in which the integrated system technology of efficient and collaborative removal of multiple pollutants including primary PM, SO₂, and NO_x was adopted to make the emission concentration of pollutants meet the strict standards. It is believed that there is limited room for greater reductions for air pollutant

Received: May 23, 2020

Revised: October 3, 2020

Accepted: October 6, 2020



ACS Publications

© XXXX American Chemical Society

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<https://dx.doi.org/10.1021/acs.est.0c03310>
Environ. Sci. Technol. XXXX, XXX, XXX–XXX

emissions²⁶ after ULERs. Given that ULERs have been basically completed in 2018, 2018 was selected as the base year for our estimation of the potential health cobenefits.

The most common method for quantifying the health effects of air pollution is the combination of a chemical transport model (CTM) and exposure–response functions.^{31–35} Nevertheless, this approach is mainly used to evaluate the health effects of an entire industry, and it is computationally infeasible to evaluate incremental perturbations of emission levels at the unit level using a CTM.³⁶ We aim at developing a unit-specific metric to address this computational burden in a manner that allows unit-level analysis.

In this study, we incorporated the unit-level health cobenefits in optimizing the early decommissioning pathway of China's coal-fired power industry. First, an emission inventory of air pollutants and CO₂ of China's power industry in 2018 was developed based on the most recent data set available. Second, the health effects of air pollution from the coal-fired power industry were evaluated by an atmospheric transport model and exposure–response functions, and an indicator called H/C was constructed to assess the relative potential health cobenefit for each unit's retirement. Third, the reliability of the H/C indicator was evaluated by comparing the results of the simulated method. Furthermore, the additional health benefits of the H/C-based strategy were estimated by comparing with other possible decommissioning strategies, and we demonstrated how the methodology could be applied for a simplified example that took different climate targets into consideration.

METHODOLOGY

Development of Coal-Fired Power Plants' Emission Inventory. A unit-based bottom-up approach was applied to develop China's coal-fired power plants' emission inventory in 2018. There were 4689 coal-fired units with a total installed capacity of 1.007 million MW (approximately 95% of the total national amount). The database was prepared from multiple sources, and the approach of the inventory development is summarized in Section S1 in the Supporting Information.

The following function was used for the core calculation of air pollution and carbon emissions.

$$E_{i,s} = U_i \times P_i \times (H_0/H_i) \times T_i \times EF_{i,s} \quad (1)$$

where i represents the coal-fired power unit number; s is the emission species; U is the installed capacity (MW); P is the coal consumption rate (kg/MWh); H_0 and H_i represent the heating values of standard coal (29.27 kJ gce⁻¹) and actual coal used in unit i , respectively; T_i is the annual utilization hours for unit i ; and EF_s is the emission factor of s pollutant (g/kg), calculated by the product of annual average pollutant concentration and theoretical flue gas volume.³⁷

Health Impact Assessment Model. Referencing previous studies,^{38–40} the integrated exposure response (IER) function⁴¹ was adopted to estimate premature deaths associated with PM_{2.5}, as a quantitative characterization of health impacts. The specific formulas are as follows:

$$M_k = \sum_m P_k \times \left(\frac{RR_m(c_k) - 1}{RR_m(c_k)} \right) \times IN_{k,m} \quad (2)$$

where k represents a specific area; m represents the specific type of disease, including stroke, ischemic heart disease (IHD), chronic obstructive pulmonary disease (COPD), and lung

cancer; M_k is the mortality associated with PM_{2.5} in area k ; P_k is the total exposed population in area; c_k is the exposed concentration in area k ; and $IN_{k,m}$ represents the baseline incidence of disease m in area k , the specific provincial value of which is sourced from the data set of Global Burden Disease (GBD) study (<http://vizhub.healthdata.org/gbd-compare/>) and other research.^{36,42} More introduction of relevant data sources can be found in Section S2 of the Supporting Information. $RR_m(c_k)$ is the relative risk for disease m at PM_{2.5} concentration c_k , which is estimated according to the following equation:

$$RR_m(c) = \begin{cases} 1, & \text{if } c < c_{0,m} \text{ otherwise} \\ 1 + \beta_{m,1} \times (1 - e^{-\beta_{m,2}(c - c_{0,m})^{\beta_{m,3}}}), & \end{cases} \quad (3)$$

where $c_{0,m}$ is the threshold concentration for disease m , which was assumed in most previous studies.^{36,43,44} $\beta_{m,1}$, $\beta_{m,2}$, and $\beta_{m,3}$ were used to describe the shapes of the exposure–response curve for disease m . The specific values of the parameters used could be found in Table S1.

The incremental health effect caused by coal-fired power plants could be calculated by different concentration inputs:

$$M_k^* = M_k(c_k) - M_k(c_{k,0})$$

where M_k^* is the premature mortality related to the emissions from coal-fired power plants, c_k is the actual PM_{2.5} concentration in the k area, and $c_{k,0}$ is the PM_{2.5} concentration when the emission from coal-fired power plants was excluded. The PM_{2.5} concentration was obtained by the weather research and forecasting model coupled with chemistry (WRF-Chem), which is a widely used tool to simulate the emission, transport, mixing, and chemical transformation of aerosols simultaneously with the meteorology.⁴⁵

Atmospheric Chemical Transport Model. In this study, the WRF-Chem model was used to simulate the PM_{2.5} concentration associated with air pollution emissions. The whole mainland of China was covered in the simulation, and the horizontal resolution was 36 km × 36 km. The employed physical parameterization includes a MOSAIC aerosol scheme,⁴⁶ Fast-J photolysis scheme,⁴⁷ and CBM-Z gas-phase chemistry.⁴⁸ The average simulation results of 4 months (January, April, July, and October) in 2018 were taken as the annual concentration, and the simulation for each month was set 5 days in advance. The anthropogenic emission inventory in 2018 was derived from the Multiresolution Emission Inventory of China (MEIC) model for mainland China and the MIX Asian inventory for perimeter regions,⁴⁹ which comprise the most updated available emission inventory data sets for China. Among the MEIC, the PM_{2.5}, SO₂, and NO_x emission inventories from the electric power sector (here we assumed it is approximately equal to the coal-fired power sector) were replaced with the unit-based inventories introduced above. Ground observation data, including meteorological conditions and air pollution concentrations, were utilized to compare the simulated results. Table S2 presents the evaluation result of the performance of WRF-Chem, and all of the values of the statistical indicators are within the benchmark ranges, reflecting the credibility of our model. More details pertaining to the configurations and evaluations are available in Section S3 of the Supporting information.

Construction of the H/C Indicator. The H/C indicator represents the health effects per amount of carbon emission, which is equivalent to the health benefit per amount of carbon reduction for each unit's retirement. The H/C indicator is directly calculated from the product of three factors in eq 4, which will be introduced in detail later, and it is normalized using eq 5 to set the national weighted average value to 1. PM_{2.5} is sometimes assumed to be linearly correlated with precursors within a small range of PM_{2.5} changes to obtain computable results, especially in policy analysis studies. Some models, such as the GAINS model⁵⁰ and Rollback model,^{15,51} which are widely used in policy research, also adopt the same assumption. Based on the linearity assumption and health impact assessment model, the relative size of the health effects caused by any two units could be represented by eq 6. Equation 7, which can be deduced from eq 6, explains the definition of the H/C indicator: it represents the ratio of the unit's proportion of the potential health cobenefit in the national total amount to the unit's proportion of carbon reduction in the national total amount. Units with H/C values greater than 1 indicate that their potential health cobenefit per amount of carbon reduction is above the national average of coal-fired power plants, and vice versa.

$$H/C_i^0 = POL_i \times POP_i \times SEN_i \quad (4)$$

$$H/C_i = H/C_i^0 \times \frac{\sum TC}{\sum (H/C^0 \times TC)} \quad (5)$$

$$\frac{Mor_i}{Mor_j} \approx \frac{TC_i \times POL_i \times POP_i \times SEN_i}{TC_j \times POL_j \times POP_j \times SEN_j} = \frac{TC_i}{TC_j} \times \frac{H/C_i}{H/C_j} \quad (6)$$

$$\frac{Mor_i}{\sum Mor} = H/C_i \times \frac{TC_i}{\sum TC} \quad (7)$$

where i or j represents specific unit i or j ; H/C_i^0 is the H/C form before the normalization of unit i ; POL is the air pollutant emissions per amount of carbon emission; POP is the affected population density in the surroundings of each unit; and SEN is the local sensitivity factor of health effects to air quality changes. Mor_i is the annual number of premature deaths caused by unit i ; TC_i is the annual total CO₂ emissions of unit i ; a more detailed introduction of the POL , POP , and SEN factors is as follows.

The POL factor represents the emission characteristics of a specific unit, i.e., the air pollution emissions per amount of carbon emissions. For coal-fired power plants, there are three main precursors that contribute to the formation of ambient PM_{2.5}, which need to be consolidated into a single metric. The POL factor is calculated as follows:

$$POL_i = \frac{(a_1 \times E_{i,SO_2} + a_2 \times E_{i,NO_x} + a_3 \times E_{i,PM}) \times \lambda_k}{E_{i,CO_2}} \quad (8)$$

where a_1 , a_2 , and a_3 represent the relative contributions of precursors to ambient PM_{2.5}, which reflects the transformation of air pollution precursors in the atmosphere. λ_k is the regional effect coefficient, which represents the regional effect of atmospheric chemical reaction on the formation of PM_{2.5}. The above-mentioned parameters are obtained from the Chinese Air Pollution Impact Model (CAPIM),¹⁸ which is an extension of the Air Pollution Emission Experiments and Policy

(APEEP) model^{52,53} in China. The air quality module in the CAPIM or APEEP models links the precursors of air pollution to ambient PM_{2.5} concentrations through the regression of historical data. The specific value of parameters can be found in Table S3.

The POP factor represents the effect of the regional population density on the amount of premature deaths. In other words, deterioration in air quality tends to have a greater health impact in more densely populated areas. Due to the complex mechanisms of PM_{2.5} formation, it is infeasible to identify the exact population affected by some certain air pollutants using a simple unified form. A land use regression (LUR) model,⁵⁴ which is an air pollution assessment model based on the principles of statistics, uses the buffer method to represent the effect of regional characteristics, such as population density, on the ambient air pollution concentration. Here, the concept of the POP factor draws on this idea, and it is defined as the population density within a buffer zone centered at the power plant location, for representing the relative size of health effects in regions with different populations. In particular, we calculated the Pearson correlation coefficient between the response concentration of the monitoring station and the surrounding air pollution emission from coal-fired power plants in Figure S1 and determined the suitable radii of the buffers by comprehensively considering the effect of short- and long-distance transmission of air pollutants. The detailed determination process of the POP factor is available in Section S4 of the Supporting Information. The average of the population density in the buffer zone with radii of 20 km and 100 km was selected as the value of the POP factor.

The SEN factor reflects the heterogeneity in health effects caused by the local PM_{2.5} concentration level and regional baseline incidence. In particular, the exposure–response function curve is not linear; thus, the sensitivity of health effects to precursor emissions varies with the local PM_{2.5} concentration level. The regional baseline incidence of the relevant disease will also affect the ultimate health outcome of ambient PM_{2.5}. Thus, we used a derivation form to represent the sensitivity of health effects to the PM_{2.5} concentration change:

$$SEN_i = \lim_{\Delta c \rightarrow 0} \frac{f_i(c_i) - f_i(c_i - \Delta c_i)}{\Delta c_i} \quad (9)$$

$$f_i(z) = \sum_m \left(\frac{RR(c_i) - 1}{RR(c_i)} \right) \times IN_{i,m}$$

where c_i and $IN_{i,m}$ represent the annual PM_{2.5} concentration and baseline incidence of disease m at unit location i , respectively. The relevant data resources are available in Section S2 of the Supporting Information.

Reliability Evaluation of the H/C Indicator. The H/C metric is a simplified tool to attribute the health effects of the electricity generation industry into each power unit. As previously mentioned, the combination of WRF-Chem and the IER function (known as the simulated method) could be used to evaluate the health effects of certain anthropogenic air emissions. In theory, we could repeatedly calculate the health effects of any concerned unit through the simulated method. However, it is difficult to individually assess the health impact of each of thousands of units due to the computational requirements of performing possibly thousands of WRF-Chem

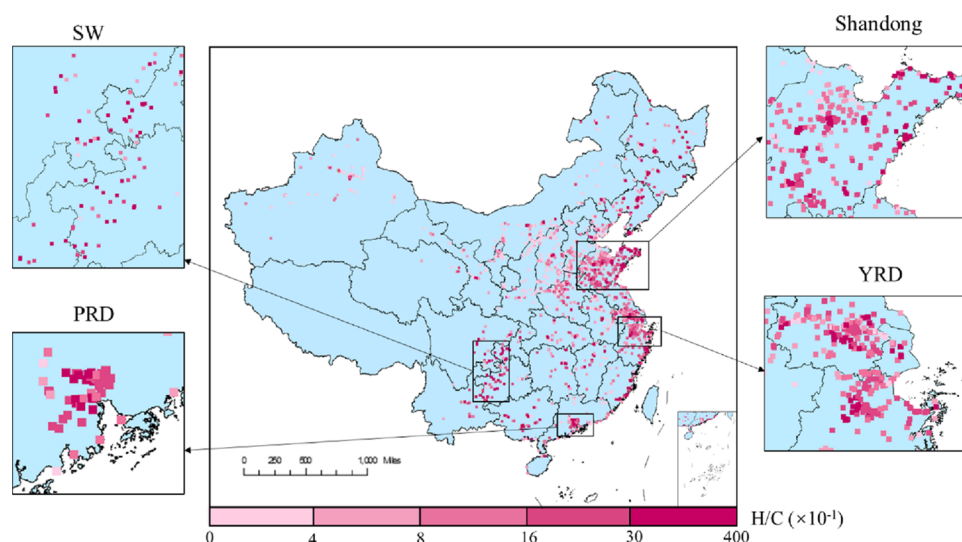


Figure 1. Spatial distribution of H/C for all of China's coal-fired power units in 2018. SW is southwest China, PRD is the Pearl River Delta, and YRD is the Yangtze River Delta.

simulations. Therefore, we designed a verification experiment to compare the results of the simulated method and H/C values for different groups of power units. In particular, all coal-fired power units were ranked using the H/C indicator in a descending order, and the coal-fired power units with 2, 5, 10, 15, 20, 30, and 50% cumulative carbon emissions were used as the inputs of the simulated method and the H/C metric to obtain comparable health estimates. The similarity of the two results was analyzed to evaluate the reliability of the H/C indicator.

H/C-Based Strategy. The rationale behind the H/C-based strategy was to assign priority to decommissioning high H/C units. We compared the health benefits of the elimination of power plants under the H/C-based strategy with two other general elimination strategies: prioritizing the elimination of older units (age-based strategy) and prioritizing the elimination of units with a smaller installed capacity (size-based strategy). A series of proportions of coal power elimination are set at percentages of 2, 5, 10, 15, 20, 30, and 50%, and the health cobenefits (avoided premature deaths each year) under the various strategies were calculated. It should be noted that the diversity in health cobenefits is fundamentally due to the differences between the units selected by each strategy. Therefore, those units that are only selected by the H/C strategy but not by the other two general strategies were also identified in this study.

The application of the H/C-based strategy was explored under global temperature target scenarios. The scenarios were established based on the analysis⁵⁵ of translating the 1.5 and 2 °C global temperature targets into the installed capacity of China's power plants to be decommissioned at different stages⁵⁵ using a version of global comprehensive assessment model for China (GCAM-China). The installed capacities of coal-fired power units that should be eliminated during the period of 2018–2030, 2030–2040, and 2040–2050 are listed in Table S4, which are derived from the previous literature.⁵⁵ A larger H/C value indicates that the corresponding power unit should be eliminated earlier. It also needs to meet the constraints of the unit life, that is, the power unit with 40 years of operation time needs to be decommissioned naturally.

RESULTS

Emissions and Health Effects of China's Coal-Fired Power Industry in 2018. The distributions of age and installed capacity among 4689 existing coal-fired power units are shown in Figures S2 and S3. The number of units with a capacity of less than 100 MW accounts for approximately 50% in the total national number, but their installed capacity accounts for only 5%. The present power generation structure in China is largely composed of 300–700 MW units.

Table S5 presents the different provincial emission factors and demonstrates that provinces have made different efforts for emission reduction in the process of implementation of ULRs. There are higher emission factors in SW regions, such as Guang Xi, Gui Zhou, and Yun Nan Provinces, approximately 2–8 times larger than those in other regions. The total national emissions from China's coal-fired power industry in 2018 is 684.8 thousand tons for SO₂, 937.9 thousand tons for NO_x, 108.7 thousand ton for PM, and 3.302 billion tons for CO₂, respectively. The spatial distributions and provincial details of the emissions are presented in Figure S4 and Table S6. Compared to the estimation in 2015 conducted by Tong et al.,²⁵ the total annual emissions of SO₂, NO₂, and PM decreased by 85, 80, and 95% in 2018, respectively, whereas the total emissions of CO₂ increased slightly.

The provincial-level disease-specific premature deaths associated with PM_{2.5} caused by coal-fired power plants emissions are listed in Table S7, and the spatial distribution of the health effects is illustrated in Figure S5. A total of 35.2 thousand premature deaths are attributed to the emission from coal-fired power plants in 2018. The specific proportions of disease attribution are 16.3% for COPD, 20.7% for IHD, 10.7% for lung cancer, and 52.3% for stroke. Moreover, Shandong (3695 premature deaths) had the largest mortality burden caused by coal-fired power plant emission, followed by Guizhou (3572 premature deaths).

H/C Values of China's Coal-Fired Power Units. The range of H/C values among China's coal-fired power units is 1.61×10^{-3} –37.7, indicating that the maximum difference of the health effect per amount of carbon emission is nearly 20,000 times. The unit with the largest H/C is located in Liaoning province with an installed capacity of 25 MW, and

that with the smallest H/C is located in Inner Mongolia with an installed capacity of 600 MW. Figure S6 presents a histogram of H/C and its logarithmic form, in which H/C exhibits a right-skewed distribution or an approximately lognormal distribution. Figure 1 presents the spatial distribution of the H/C value for each unit and clearly indicates a spatial aggregation pattern. A detailed view of four high-value aggregation areas is obtained: Southwest (SW) China, the Pearl River Delta (PRD), the Yangtze River Delta (YRD), and Shandong Province. The PRD and YRD are well-known economic regions in China, and their concentrated population levels imply enormous health benefits from the coabatement of air pollution. The reason for higher H/C values in SW China is caused by the higher air pollution emission factors of power plants there. Shandong province is similarly affected by these two factors. Beijing-Tianjin-Hebei (BTH) is also a well-known economic region and a high-profile area for air pollution research, whereas its H/C value is not relatively high, which is likely due to the stringent environmental actions occurred in these areas with stricter emission standards. The low emission levels and fewer unit numbers result in limited health benefits of carbon reduction in the BTH region.

Figure 2 presents the distribution of H/C and the cumulative health cobenefits associated with cumulative

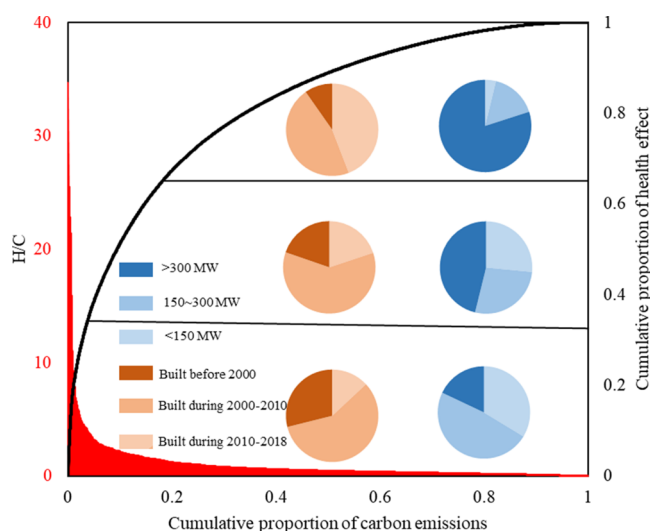


Figure 2. H/C value and its cumulative distribution of national power plant units. Red: The H/C histogram of each unit in the descending order of their H/C values. Black curve: Cumulative potential health cobenefits corresponding to the units with the cumulative carbon emissions. Pie chart: Distribution of years in which power plants were built and their installed capacities divided into three categories based on the health effect trisection. The pie chart area represents the proportion of CO₂ emissions.

carbon emission. A nonlinear relationship exists between the cumulative health effect and cumulative carbon emission. In particular, the first 1, 5, and 20% of carbon emission could contribute to approximately 20, 40, and 65% of the total health effect. In other words, considerable health benefits are solely generated from the coabatement air pollution of a limited number of power units. Since health benefits are a significant driving force for promoting the implementation of China's climate policy in the short term, a more efficient strategy that can quickly achieve greater health benefits is required. To analyze the H/C composition, all units were divided into three

groups based on the trisection of health effects. Of all the units, 11.6, 37.2, and 51.2% were included in the three groups, accounting for 3.8, 15.7, and 80.5%, respectively, of the carbon emissions. It can be viewed on the pie chart that the group with a higher H/C value is composed of older and smaller units.

In terms of the economic cost incurred in the elimination, eliminating a newer unit will have a larger price of stranded assets than an older one. Similarly, the cost of eliminating some larger units with expensive control equipment may be higher than that of eliminating smaller ones. It means that the economic cost of eliminating older or smaller units will be lower but higher health benefits will be obtained. Figure S7 exhibits the relationship between the cumulative proportion of the unit quantity and cumulative proportion of the health effect. The heterogeneity degree on the unit quantity curve is smaller than that of carbon emissions. For example, 20% of carbon emission reductions contributed to 70% of the total potential health cobenefits, corresponding to 56% of all units. This is because smaller units are more likely to have larger H/C values.

The reliability of the H/C metric was evaluated by comparing the results from H/C values (proxy values) with the results of the simulated method (simulated values) for different groups of power units, and the performance is presented in Figure 3. For different inputs of power plants, the

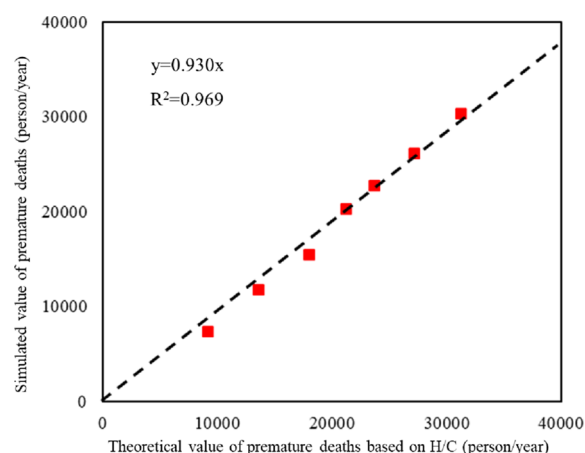


Figure 3. Reliability evaluation of the H/C metric. Simulated value by the simulated method vs proxy value calculated from the H/C metric of premature deaths associated with PM_{2.5}.

simulated and proxy values of health effects reflect good consistency (slope = 0.93, $R^2 = 0.969$, when the forced intercept is 0). A slope of less than 1 indicates that the overall proxy value is relatively high. The selected group in this experiment consists of units with the highest H/C values, and the extreme values are more likely to have an estimation bias. In general, the health effect results using the H/C metric are close to those calculated using the traditional method with a high computational complexity. Therefore, the H/C metric could be used as a reliable tool to analyze the health effects at the unit level.

Additional Health Benefits of the H/C Based Strategy. Figure 4 presents the results of avoided premature deaths for three elimination strategies under seven scenarios. In the 2% retirement scenario, the health benefits are 2.29, 7.42, and 1.69 thousand avoided premature deaths for the size-

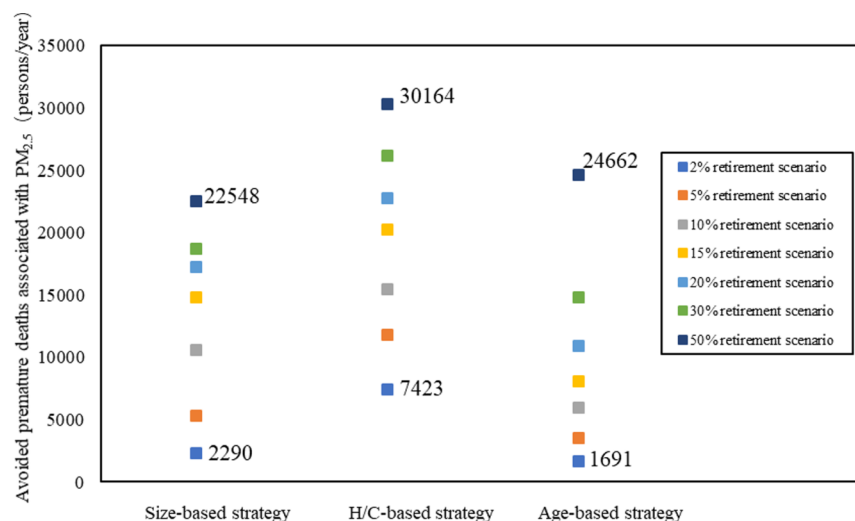


Figure 4. Comparison of avoided premature deaths for three elimination strategies under seven scenarios characterized by different retirement scenarios.

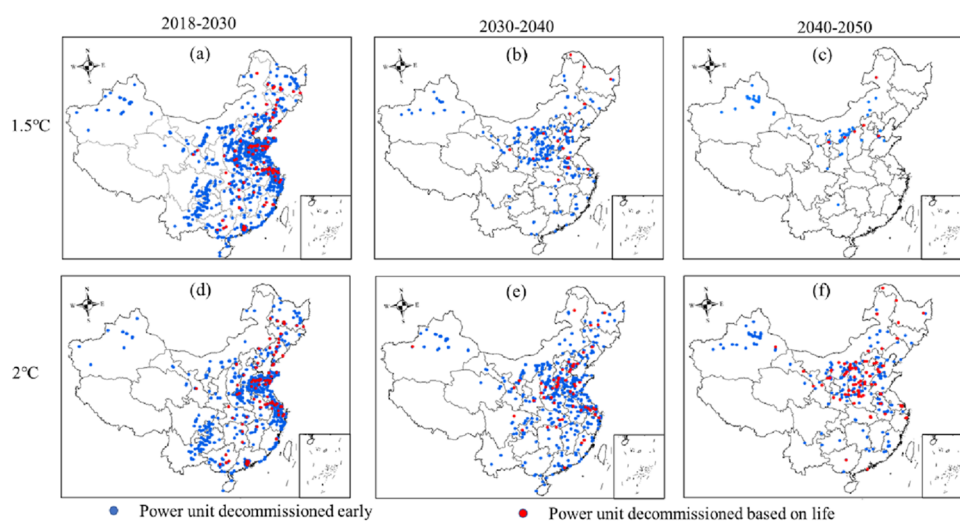


Figure 5. Maps of units to be eliminated in each stage under climate target scenarios: (a) 2018–2030 under 1.5 °C scenario; (b) 2030–2040 under 1.5 °C scenario; (c) 2040–2050 under 1.5 °C scenario; (d) 2018–2030 under 2 °C scenario; (e) 2030–2040 under 2 °C scenario; and (f) 2040–2050 under 2 °C scenario.

based strategy, H/C-based strategy, and age-based strategy, respectively, and in the 50% retirement scenarios, the corresponding results are 22.5, 30.2, and 24.7 thousand avoided premature deaths, respectively. Compared to the size-based strategy and age-based strategy, maximum additional health benefit of the H/C strategy can reach 12.1 and 7.76 thousand avoided premature deaths in 15 and 50% retirement scenarios, respectively. Comparing the size-based and age-based strategy, the former has greater health benefits under the scenario of a low proportion of retirement, whereas the latter has greater health benefits under the higher retirement scenarios. Although the H/C-based strategy inherently optimizes health benefits, the fractional benefit relative to other strategies diminishes as more coal plants are replaced. Therefore, it is worth noting that the benefit of optimizing decision-making based on the H/C indicator may be very limited in scenarios in which the government closes a large number of coal-fired plants (more than 50%) in a short time period. However, given China's coal-dominated energy

structure and the demand for stable power generation stability, the aforementioned situation is very unlikely to occur in China.

The difference in health benefits among the analyzed strategies is generated from their concerned units. Some units that are selected by the H/C-based strategy but not selected by other general strategies actually have a great potential for health cobenefits, but do not get enough attention in general. The spatial locations of those key units that are a part of the H/C-based strategy are mapped in Figure S8. There are approximately 106–180 related units in each scenario. Under different emission reduction targets, the units that require additional attention are not completely consistent, but present a spatial cluster distribution. For example, a large number of units in the eastern coastal or southwestern regions that are not considered under the general elimination strategy do comprise a large proportion of health benefits, due to proximity to dense population or high air pollution emission rates.

Application of the H/C Indicator. The H/C-based strategy is applied to the planning of the elimination pathway

of China's coal-fired power industry. Figure 5 presents the distribution of units that should be decommissioned at each stage under the 1.5 and 2 °C scenarios. Some of the units (indicated by the red dot) will be normally eliminated when reaching the unit age of 40 years, and other units (indicated by the blue dot) will be artificially scheduled to be decommissioned before the unit age of 40 years due to climate constraints. This study is to optimize the retirement priority of blue units to obtain more health benefits. In the six figures, the proportion of blue units is obviously greater than that of red units, indicating that only a small number of units will be naturally decommissioned in our policy scenario. The 1.5 °C scenario has higher requirements on the schedule of the power unit elimination, and a larger proportion of power units need to be eliminated during the period of 2018–2030. Between the two scenarios, the elimination priority of coal-fired power units presents similar characteristics. The first batch of decommissioned coal-fired power units under the two scenarios is concentrated in SW China, North China, and eastern coastal areas. The last batch of decommissioned power units is mainly located in northwest and central China. Due to the late built time of these power units and the relatively low health benefits obtained from the decommissioning process, they could be considered at the end of the period to implement low-carbon measures. Figure S9 presents the power units to be eliminated under the H/C-based strategy but not under size-based or age-based strategies, which reflects the influence of the H/C metric on the selection of which units to decommission. According to the number of affected units in two scenarios, this influence is greater in the 2 °C scenario than in the 1.5 °C scenario, and it indicates that the optimization space of the H/C metric for decommissioning of power units will decrease with an increase in the elimination strength.

DISCUSSION

The results of our analysis reveal that the H/C metric is distributed very unevenly across power units, and significant health benefits can be generated following the H/C-based strategy. By definition, H/C is completely determined by the POP, POL, and SEN factors, and our analysis shows that the H/C value tends to be higher at older and smaller power plants. To explore the internal relationship among these factors, the correlation coefficients of the six variables are listed in Table S8. Among the three determinant factors, H/C is more correlated with nearby population density, followed by air pollution emission rates and the local sensitivity factor. Moreover, the correlation depends on the relative size of the coefficient of variation (POP:0.970, POL:0.652, SEN:0.415). The uneven distribution of the nearby population in different regions of China is the prime factor affecting H/C, which is consistent with our previous research at the power grid level.¹⁵ As an independent influence on H/C, the air pollution emission factor also determines the H/C to a large extent, resulting in a higher H/C value for some units in areas with relatively smaller populations but higher air pollutant emission rates per unit of carbon emission. The local sensitivity factor is negatively correlated with the nearby population density, which weakens the influence of the nearby population density on the H/C due to the high level of medical care with a lower baseline incidence of PM_{2.5}-related diseases in densely populated areas. In addition, the correlation analysis results reveal that replacing those smaller and older units would have a relatively higher potential of health benefits, as they tend to

have higher air pollution emission rates (due to the absence of advanced control equipment) or be located in more densely populated locations.

The present study has many limitations and areas that can be improved in the future. We only compared the aggregate results for a handful of scenarios to discuss the reliability of the H/C. It is computationally difficult for the conventional simulation method to individually simulate perturbations to the emissions of thousands of units. Under the condition of limited computing resources, a series of scenarios in which different proportions of coal-fired power plants are phased out are set to compare the health results between H/C values and the simulated method. More verification schemes are needed to fully determine the reliability of the H/C indicator. In the development of H/C metric, we have simplified some practical factors such as the downwind effect. PM_{2.5} impacts do not happen only in a uniform radius around a power plant but instead depend on the flow of the plume downwind. In addition, the relationship between PM_{2.5} and its precursors is not strictly linear. These simplified assumptions will lead to the deviation of the H/C method, which is reflected in the difference between the proxy value and the simulated value. In terms of policy analysis, the present research also has some areas for improvement. We only focused on the process of power generation and neglected the impact of other energy processes such as power transmission. Moreover, the basic unit chosen for the study is per amount of carbon reduction, while the economic cost per carbon reduction among power plants is different, depending on the power generation efficiency or operational year. These limitations could be improved by coupling with energy system models or economic analysis.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.0c03310>.

Development approach and result of emission inventory; relevant data source; configuration, evaluation, and performance of WRF-Chem; determination process of the POP factor; values of parameters used in the POL factor, IER function, and scenario setting; results of premature mortality; distribution of H/C values; and spatial distribution of key power units (PDF)

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (No. 71773062 and No. 71525007) and the National Social Science Foundation of China (No.17ZDA077).

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